

RADICAL – Radiology AI-Driven Clinical Decision-making with Multi-modal Exploration

The rapid digital transformation in healthcare has led to an unprecedented increase in both the volume and variety of data generated. This data explosion overwhelms clinicians, as it often exceeds the human capacity to process and analyze data effectively. Much of this information is stored in unstructured formats, such as text and images, making it difficult to interact with and use in an integrated and intelligent way. As a result, there is a need for advanced tools that can explore, extract, connect, and represent health and personal data for clinical decision-making processes.

Radiology, a cornerstone of medical diagnostics and patient management, faces significant challenges related to fragmented data and incomplete information. Radiologists often struggle to make informed clinical decisions due to the disjointed nature of medical records, which consist of various sources such as human-written texts, lab data, radiology and pathology reports, medication plans, and medical images. These records are typically presented in chronological order and mostly consist of plain text with timestamps, making them difficult to search and analyze efficiently. Additionally, for legal and regulatory reasons, the original medical records must remain unaltered, further complicating the process of information extraction, and leading to repetitive and time-consuming manual effort.

These data-related challenges not only hinder radiologists but also healthcare professionals across the board, who spend a significant amount of time reviewing patient information for daily practice. The lack of a cohesive and easily searchable system can delay critical clinical decisions and follow-up care, potentially impacting patient outcomes.

To address these issues, we propose the development of **RADICAL** to facilitate access to various types of medical information that exist in hospital systems and assist radiology professionals in extracting relevant data from a patient's medical record based on natural language queries. This tool will streamline the process of writing radiology reports, enable retrospective research, and ensure that important findings receive timely follow-up, ultimately improving healthcare quality and patient outcomes. Our proposed system will act as a medical AI assistant capable of analyzing complex medical records while maintaining the integrity of the original data, thereby enhancing the efficiency of clinical practice without compromising legal or regulatory requirements. Such advanced data exploration tools have the potential to enhance radiology and transform broader clinical practice. In clinical settings, decision-making often requires multi-step reasoning, careful planning, and interaction with various data sources to uncover meaningful insights. Enabling seamless integration of diverse data types will empower clinicians to make more informed, personalized decisions, improve patient care, and ensure that critical findings are addressed promptly across specialties.

A. Context and Problem Description

Radiologists and healthcare providers encounter various challenges in their daily practice due to the limitations of current electronic health systems. Some of these challenges are illustrated by the following use case as depicted in Figure 1, where “a radiologist aims to provide a radiology report for a computer tomography (CT) scan”.

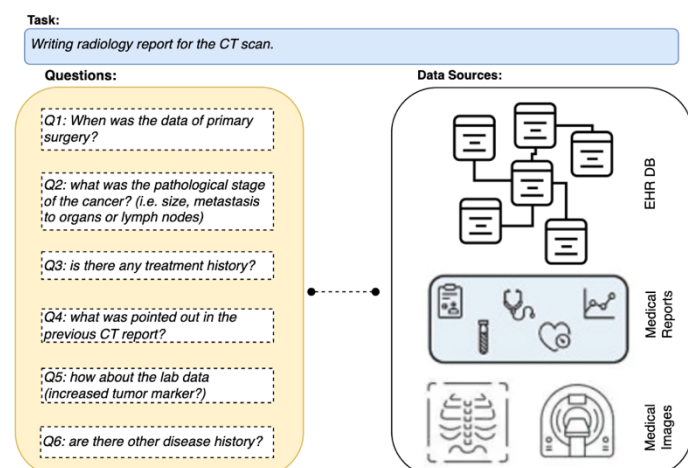


Figure 1: Example of a context-intensive radiological task. Writing follow-up CT reports for complex oncological cases usually involves retrieving details about the date and kind of surgery performed (Q1), the diagnosed oncological entity and stage (Q2), other non-surgical treatments received (Q3), the findings reported for the previous CT study (Q4), any currently relevant laboratory values (for instance, inflammation markers to narrow down the differential diagnoses; Q5), and a scoping review of other potentially relevant history (Q6).

Radiologists often receive CT scan orders with vague or insufficient information. Comments such as "Patient presents with abdominal pain" or "Evaluate the effectiveness of chemotherapy" lack the clinical details needed for accurate image interpretation. This absence of specificity requires radiologists to search multiple data sources manually—Electronic Health Records (EHR), lab results, pathology reports, and previous imaging records—to gather the necessary information. This process is time-consuming, delays reporting, and may lead to incomplete diagnoses, affecting patient care, while fragmented healthcare data systems further complicate effective follow-up tracking.

To prevent such oversights, there is a need for a system that can consolidate all relevant data from different sources into a single, accessible platform. To address this, several medical AI systems that process a wide range of data inputs have been introduced recently, including models that integrate radiology images with clinical data [17, 18].

However, they focus on performing only one task at a time and are limited to narrow applications. Alongside these, multi-purpose AI models—especially Foundation Models—are gaining traction in healthcare. Examples include Med-PaLM [19], LLaVa-Med [20], MEDITRON [21], and MAIRA-2 [22] which leverage large datasets to handle diverse medical tasks. However, these models face challenges in updating evolving medical knowledge and managing a distribution shift, where model performance deteriorates as input data or medical practices change.

Another solution to these challenges involves Retrieval-Augmented Generation (RAG) and instructed-tuning approaches, which augment Foundation Models with real-time, external data to better handle specific medical questions [23]. Yet, a trade-off exists: while these models are versatile, they lack specialization and expertise in individual tasks. This all-in-one approach often results in diminished accuracy in critical areas, as no single model can excel at everything. Ideally, instead of relying on a heavy, monolithic Foundation Model, the future of medical AI may lie in **agentic AI systems**—models capable of autonomously managing tasks, continuously adapting, and learning from real-world environments [24]. These systems can specialize in individual medical tasks while still integrating seamlessly into broader healthcare frameworks. Outside of medicine, agentic models like Caesura [25] and innovations from big tech companies [26, 27] have demonstrated the potential of adaptable AI architectures. Bringing these principles into healthcare could pave the way for dynamic, task-oriented AI solutions that address the limitations of all-in-one models. This approach is particularly important in medical decision-making, which often requires multi-step reasoning, careful planning, and interaction with various data sources to uncover meaningful insights.

The most recent AI-agent driven method in the medical domain is MedVersa [28], a versatile learner that facilitates adaptable learning and multitasking in medical image interpretation by leveraging a large language model as an orchestrator. However, their focus is restricted to image and text modalities, with limited model explainability. Additionally, their scope is confined to medical image interpretation, while we aim to address a broader range of tasks involving diverse medical data exploration and discovery.

B. Proposed Method and Implementation Strategy

Our proposed method is an AI agent-based system, as depicted in Figure 2, composed of several key components that work together to enable autonomous task completion to respond to user questions. The anatomy of this agentic AI system consists of the following main elements: i) **LLM- The brain:** At the core of our system is a Large Language Model (LLM) acting as the agent's "brain," providing intelligence, language understanding, and reasoning. It enables natural language interactions, context comprehension, tool orchestration, and decision-making, either directly interacting with or enabling the system's components. ii) **Planning - Decomposing a user question into tasks:** A crucial capability of our system is the ability to plan, which allows it to break down complex user queries into manageable tasks, analyzing intent and prioritizing steps logically. This adaptive process, supported by LLMs for reasoning, is essential for efficiently solving multi-step problems. iii) **Action - Use of tools:** The action capacity enables our system to engage with its environment by using data and tools, such as performing analysis, accessing databases, or running code. This expands the system's task range, shifting from generative processes to deterministic execution. iv) **Profile - Specify System Behavior:** The system's profile defines its behavior, knowledge, and abilities, including role, tone, ethics, and expertise. This ensures consistent and appropriate actions aligned with its intended function.

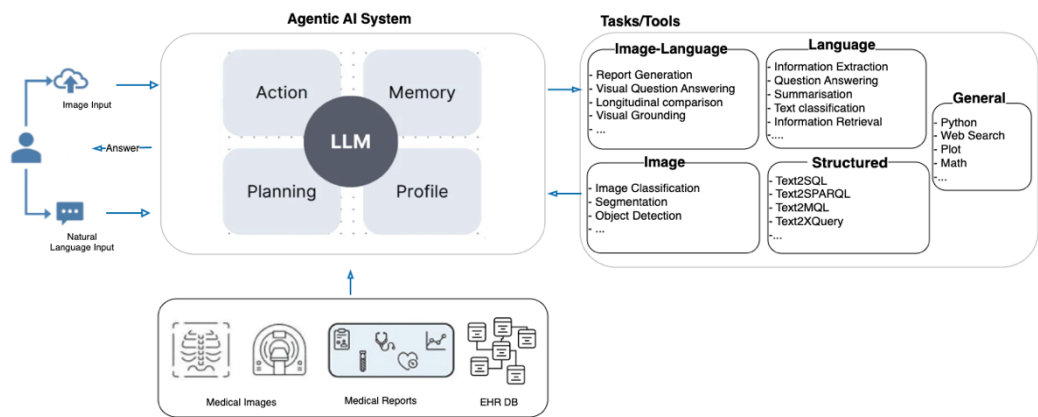


Figure 2: Our proposed system (**RADICAL**) using agentic concepts: The system can take autonomous actions and decisions with minimal human intervention to answer complex natural language questions over tabular data, text, and images. The system decides which tools to choose for analyzing the data, e.g. Text2Query for structured data or image analysis, and intelligently combines the results. At each stage, the answers are explained by the system in natural language and humans could verify the results. It thus overcomes the major hallucination problem of current LLMs that are often considered black boxes.

v) Memory - Storing and recalling past experiences: The memory function allows the system to retain and retrieve information from past interactions, maintaining context, learning from experiences, and providing personalized responses. It can be short-term for single conversations or long-term across multiple interactions. vi) Environment: In our AI-driven ecosystem, the environment in which the system operates is defined by several key components: a *data lake*, *tools*, and *human feedback*. *Data lake* stores extensive structured and unstructured data, providing the system with a broad information base for analysis and decision-making. This repository enables the extraction of insights, pattern identification, and generation of informed responses. *Tools* include APIs, software applications, and external platforms that allow the system to perform specific tasks like running code or accessing databases. They shift the system’s function from generative processes to deterministic actions based on real-time requirements. *Human feedback* helps refine the system’s performance by allowing it to learn from errors and adjust its behavior. This feedback loop ensures that the system remains relevant, accurate, and adaptable to new challenges. Together, these components—data lake, tools, and human feedback—form an integrated environment that enables our system to operate efficiently, learn continuously, and improve its ability to meet complex, evolving tasks.

We will employ LangGraph¹ to implement **RADICAL**. LangGraph integrates with both open-source and closed-source large language models (LLMs), but for real-data applications, we prioritize open-source LLMs due to their transparency, community support, and adaptability. Open-source models also address privacy concerns more effectively, allowing for greater control over data handling and security practices. This approach aligns with our commitment to open development and ensures robust, flexible, and privacy-conscious integrations with real medical data.

C. Collaboration Schema and Work packages

Our collaboration forms a strong synergy between multiple DIZH institutions as follows:

1. Intelligent Information Systems Group (ZHAW): The Intelligent Information Systems Group (IIS) at ZHAW plays a pivotal role, providing expertise in the design and implementation of the proposed framework. The main research focus of the IIS group lies at the intersection of information systems, natural language processing, and machine learning. Within our collaboration, IIS is represented by Dr. Farhad Nooralzadeh, the Principal Investigator (PI) of this project. His research interests center around computational approaches to language understanding, the integration of multimodal and multilingual aspects in machine learning, and the diverse applications of Generative AI particularly Large Language Models across various

¹ <https://langchain-ai.github.io/langgraph/>

domains. His recent related works include Bilingual Open Data Exploration in Natural Language [1], Improving the Cross-Lingual Generalization in Visual Question Answering [2], Radiology Report Generation Models Evaluation Dataset for Chest X-rays [3], and Progressive Transformer-Based Generation of Radiology Reports [4], all of which reflect his expertise in bridging computational models with practical applications, particularly in multilingual, multimodal, and healthcare-focused AI systems.

2. Chair of Medical Informatics (UZH): The Chair of Medical Informatics is affiliated with the University Hospital of Zurich (USZ) and the Department of Quantitative Biomedicine at the University of Zurich (UZH). As such, the lab is intimately involved in addressing the data processing needs of the Zurich health system. Currently, they are leading major digital medicine initiatives² in the fields of cancer, rheumatology, and intensive care medicine. The group is also leading major data integration efforts between the Zurich hospitals and is developing advanced data-sharing methodologies³. The lab has major experience with working with clinical datasets for performing NLP [11, 12, 13], medical image processing [14], and outcome prediction [15]. A major focus is the use of AI in clinical practice. Particularly, they are developing AI methods that enable physicians to effectively study patient trajectories for treatment selection and care pathway discovery [16]. This group is represented by *Prof. Dr. Michael Krauthammer*, who serves as the department director, he is a trained physician and computer scientist with long-lasting experience in ML for healthcare.

3. Institute for Diagnostic and Interventional Radiology (USZ): The Institute for Diagnostic and Interventional Radiology at the University Hospital of Zurich (USZ), as one of the largest radiology institutes in Switzerland, is actively involved in the research, translation, and clinical application of modern medical imaging and associated support processes. USZ as a practice partner is represented by Dr. med. *Christian Blüthgen*. Dr. Blüthgen is a board-certified radiologist specializing in thoracic imaging and is currently an attending radiologist at the Institute for Diagnostic and Interventional Radiology at USZ. He completed a postdoctoral fellowship in an interdisciplinary (radiologists, clinicians, computer scientists) research group (Stanford Center for Artificial Intelligence in Medicine and Imaging (AIMI)) and has extensive hands-on experience in all aspects of developing multimodal AI systems. His previous contributions and projects at the intersection of AI and healthcare included comparing the performance of adapted large language models and clinicians for clinical text summarization [5], the development of a generative model capable of creating synthetic chest X-ray images [6], a foundation model for abdominal CT analysis [7], several works using deep learning-based approaches for fracture and aneurysm detection [8, 9], and radiomics-based characterization of thymic epithelial tumors [10].

The ZHAW group, in close collaboration with the Chair of Medical Informatics group from UZH, the MDS group⁴ from ETH, and the University Hospital Zurich (USZ), is working on the development of multimodal chest X-ray (CXR) foundation models as part of the Swiss AI Initiative⁵ - Health. This collaboration leverages the Swiss AI infrastructure to meet the high computational demands essential for this next generation of health foundation models.

This multidisciplinary team highlights the power of collaboration between various institutions and fields of study. In addition, our close collaboration with the practice partner (USZ) guarantees that the solutions we develop are both grounded in solid theory and practical for real-world radiology applications, leading to measurable improvements in service delivery. Further details about the team members can be found in the attached CVs.

An overview of the work packages is given in Table below. The details of each work package are explained below.

Work Package	Year 1			Year 2			Year 3		
WP1: Data and tools set curation									
WP2: Design and Implementation									

² <https://dizh.ch/en/2022/07/07/zurich-applied-digital-health-center/>

³ <https://theloopzurich.ch/en/focal-points/the-loop-zuerich-biomedical-informatics-platform/>

⁴ <https://mds.inf.ethz.ch/>

⁵ <https://www.swiss-ai.org/>

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